

Rik Koncel-Kedziorski

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[Google Scholar](#)

Research Interests

My research focuses on new methods for reasoning from data, human-computer collaboration, and improving evaluation of AI systems. Previously, I worked on those same things plus question answering, language grounding, and pragmatic meaning representation.

Education

University of Washington 2019

PhD, Computational Linguistics

Dissertation: Understanding and Generating Multi-sentence Texts. Advisors: Hanna Hajishirzi & Gina-Anne Levow.

The Evergreen State College 2012

B.S./B.A.

Miami-Dade College 2009

A.A.

Academic Positions

University of Washington, Paul G. Allen School of Computer Science & Engineering

Postdoctoral Researcher

2019–2021

Research on language understanding and generation for scientific text, human-AI interaction, and dialog systems.

PI: Noah A. Smith.

Industry Positions

Apple

ML Scientist

2024–present

Research on personalization, preference learning, and reasoning for multimodal LLM systems; synthetic data generation for user-adaptive models.

S&P Global, Kensho

Research Scientist

2022–2024

Research on financial NLP and reasoning; datasets and evaluation methods for LLM-based financial question answering.

Amazon

Applied Scientist

2021–2022

Generative question answering for Alexa; retrieval-augmented generation and multilingual QA.

Publications

Peer-Reviewed

- [1] Charles Lovering, Michael Krumdick, V. Lai, Varshini Reddy, Seth Ebner, Nilesh Kumar, **Rik Koncel-Kedziorski**, Chris Tanner. “Language Model Probabilities are Not Calibrated in Numeric Contexts.” *Annual Meeting of the Association for Computational Linguistics*, 2025.
- [2] Brihi Joshi, Xiang Ren, Swabha Swayamdipta, **Rik Koncel-Kedziorski**, Tim Paek. “Improving Language Model Personas via Rationalization with Psychological Scaffolds.” *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [3] **Rik Koncel-Kedziorski**, Brihi Joshi, Tim Paek. “PrimeX: A Dataset of Worldview, Opinion, and Explanation.” *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [4] Varshini Reddy, **Rik Koncel-Kedziorski**, V. Lai, Michael Krumdick, Chris Tanner. “DocFinQA: A Long-Context Financial Reasoning Dataset.” *Annual Meeting of the Association for Computational Linguistics*, 2024.
- [5] **Rik Koncel-Kedziorski**, Michael Krumdick, V. Lai, Varshini Reddy, Charles Lovering, Chris Tanner. “BizBench: A Quantitative Reasoning Benchmark for Business and Finance.” *Annual Meeting of the Association for Computational Linguistics*, 2024.

- [6] Matteo Gabburo, Siddhant Garg, **Rik Koncel-Kedziorski**, Alessandro Moschitti. “Learning Answer Generation using Supervision from Automatic Question Answering Evaluators.” *Annual Meeting of the Association for Computational Linguistics*, 2023.
- [7] Yao Dou, Maxwell Forbes, **Rik Koncel-Kedziorski**, Noah A. Smith, Yejin Choi. “Is GPT-3 Text Indistinguishable from Human Text? Scarecrow: A Framework for Scrutinizing Machine Text.” *Annual Meeting of the Association for Computational Linguistics*, 2022.
- [8] Matteo Gabburo, **Rik Koncel-Kedziorski**, Siddhant Garg, Luca Soldaini, Alessandro Moschitti. “Knowledge Transfer from Answer Ranking to Answer Generation.” *Conference on Empirical Methods in Natural Language Processing*, 2022.
- [9] Benjamin Muller, Luca Soldaini, **Rik Koncel-Kedziorski**, Eric Lind, Alessandro Moschitti. “Cross-Lingual Open-Domain Question Answering with Answer Sentence Generation.” *AACL*, 2022.
- [10] Zhilin Wang, Xuhui Zhou, **Rik Koncel-Kedziorski**, Alex Marin, Fei Xia. “Extracting and Inferring Personal Attributes from Dialogue.” *NLP4CONVAI*, 2022.
- [11] Zeqiu Wu, Michel Galley, Chris Brockett, Yizhe Zhang, Xiang Gao, Chris Quirk, **Rik Koncel-Kedziorski**, Jianfeng Gao, Hannaneh Hajishirzi, Mari Ostendorf, Bill Dolan. “A Controllable Model of Grounded Response Generation.” *AAAI Conference on Artificial Intelligence*, 2021.
- [12] Kelvin Luu, Xinyi Wu, **Rik Koncel-Kedziorski**, Kyle Lo, Isabel Cachola, Noah A. Smith. “Explaining Relationships Between Scientific Documents.” *Annual Meeting of the Association for Computational Linguistics*, 2021.
- [13] Sachin Mehta, **Rik Koncel-Kedziorski**, Mohammad Rastegari, Hannaneh Hajishirzi. “DeFINE: Deep Factorized Input Token Embeddings for Neural Sequence Modeling.” *International Conference on Learning Representations*, 2020.
- [14] Mark Hopkins, Ronan Le Bras, Cristian Petrescu-Prahova, Gabriel Stanovsky, Hannaneh Hajishirzi, **Rik Koncel-Kedziorski**. “SemEval-2019 Task 10: Math Question Answering.” *International Workshop on Semantic Evaluation*, 2019.
- [15] **Rik Koncel-Kedziorski**, Dhanush Bekal, Yi Luan, Mirella Lapata, Hannaneh Hajishirzi. “Text Generation from Knowledge Graphs with Graph Transformers.” *North American Chapter of the Association for Computational Linguistics*, 2019.
- [16] Aida Amini, Saadia Gabriel, Shanchuan Lin, **Rik Koncel-Kedziorski**, Yejin Choi, Hannaneh Hajishirzi. “MathQA: Towards Interpretable Math Word Problem Solving with Operation-Based Formalisms.” *North American Chapter of the Association for Computational Linguistics*, 2019.
- [17] Sachin Mehta, **Rik Koncel-Kedziorski**, Mohammad Rastegari, Hannaneh Hajishirzi. “Pyramidal Recurrent Unit for Language Modeling.” *Conference on Empirical Methods in Natural Language Processing*, 2018.
- [18] **Rik Koncel-Kedziorski**, Subhro Roy, Aida Amini, Nate Kushman, Hannaneh Hajishirzi. “MAWPS: A Math Word Problem Repository.” *North American Chapter of the Association for Computational Linguistics*, 2016.
- [19] Aaron Jaech, **Rik Koncel-Kedziorski**, Mari Ostendorf. “Phonological Pun-derstanding.” *North American Chapter of the Association for Computational Linguistics*, 2016.
- [20] Kirby Conrod, Rachael Tatman, **Rik Koncel-Kedziorski**. “We Who Tweet: Pronominal Relative Clauses on Twitter.” *Corpus Linguistics Fest*, 2016.
- [21] **Rik Koncel-Kedziorski**, Ioannis Konstas, Luke Zettlemoyer, Hannaneh Hajishirzi. “A Theme-Rewriting Approach for Generating Algebra Word Problems.” *Conference on Empirical Methods in Natural Language Processing*, 2016.
- [22] **Rik Koncel-Kedziorski**, Hannaneh Hajishirzi, Ashish Sabharwal, Oren Etzioni, S. Ang. “Parsing Algebraic Word Problems into Equations.” *Transactions of the Association for Computational Linguistics*, 2015.
- [23] **Rik Koncel-Kedziorski**, Hannaneh Hajishirzi, Ali Farhadi. “Multi-Resolution Language Grounding with Weak Supervision.” *Conference on Empirical Methods in Natural Language Processing*, 2014.

Preprints

- [24] **Rik Koncel-Kedziorski**, Noah A. Smith. “Go Forth and Prosper: Language Modeling with Ancient Textual History.” *arXiv*, 2021.
- [25] Zeqiu Wu, **Rik Koncel-Kedziorski**, Mari Ostendorf, Hannaneh Hajishirzi. “Extracting Summary Knowledge Graphs from Long Documents.” *arXiv*, 2020.
- [26] Benjamin Robaidek, **Rik Koncel-Kedziorski**, Hannaneh Hajishirzi. “Data-Driven Methods for Solving Algebra Word Problems.” *arXiv*, 2018.

Advising

Graduate Students

- **Brihi Joshi**, USC — PhD Student (Apple Scholar) *2024–present*
- **Kelvin Luu**, UW CSE — PhD Student *2019–2021*
- **Ellen Wu**, UW ECE — PhD Student *2019–2021*
- **Kyle Yan**, UW CSE — Masters Student *2019–2021*
- **Xinyi Wu**, UW Linguistics — Masters Student *2019–2021*
- **Xuhui Zhou**, UW Linguistics — Masters Student *2019–2021*

Undergraduate Researchers

- **Yao Dou**, UW CSE *2019–2021*
- **Velocity Yu**, UW CSE *2019–2021*
- **Annabell Hu**, UW CSE *2019–2021*
- **Daniel Cheng**, UW CSE *2019–2021*

Research Interns

- **Nilesh Kumar**, Kensho *2022–2024*
- **Matteo Gabburo**, Amazon *2021–2022*
- **Benjamin Muller**, Amazon *2021–2022*

Grants and Awards

Apple AI Research Grant *2024*
"User Representations and Reasoning for Natural Dialogue Systems", in collaboration with USC Professor Swabha Swayamdipta

Postdoc Research Award *2020*
"Promoting Crowd-participation in Dialog System Research", University of Washington Computer Science

Excellence in Linguistic Research Fellowship *2018*
University of Washington Linguistics

STEM Scholarship *2010, 2011*
National Science Foundation

Professional Service

Area Chair: ACL Rolling Review (ARR)

Reviewing: *ACL, ICLR, NeurIPS, WWW, AAAI; WideningNLP and other workshops focused on improving access to research; Apple Scholars fellowship program (industrial-academic)

Contributions: OECD AI and the Future of Skills (2023); SemEval 2019 Task 10: Math Question Answering (organizer)

Teaching

Instruction

- *Instructor*, UW: Ling 200: Introduction to Linguistic Thought
- *Teaching Assistant*, UW: CSE 373: Data Structures and Algorithms
- *Teaching Assistant*, UW: Ling 200: Introduction to Linguistic Thought
- *Teaching Assistant*, Evergreen State College: Graphic Programming with Processing

Guest Lectures

- "Language Models" in CSE 573: Artificial Intelligence
- "Machine Translation" in UW Translation Hub Lecture Series
- "Deep Learning in Computational Linguistics" in Ling 472: Introduction to Computational Linguistics

- “Text Generation” in CSE 473: Introduction to Artificial Intelligence